External Gateway Routing Protocols: BGP & MP-BGP

Redes de Comunicações II

Licenciatura em Engenharia de Computadores e Informática

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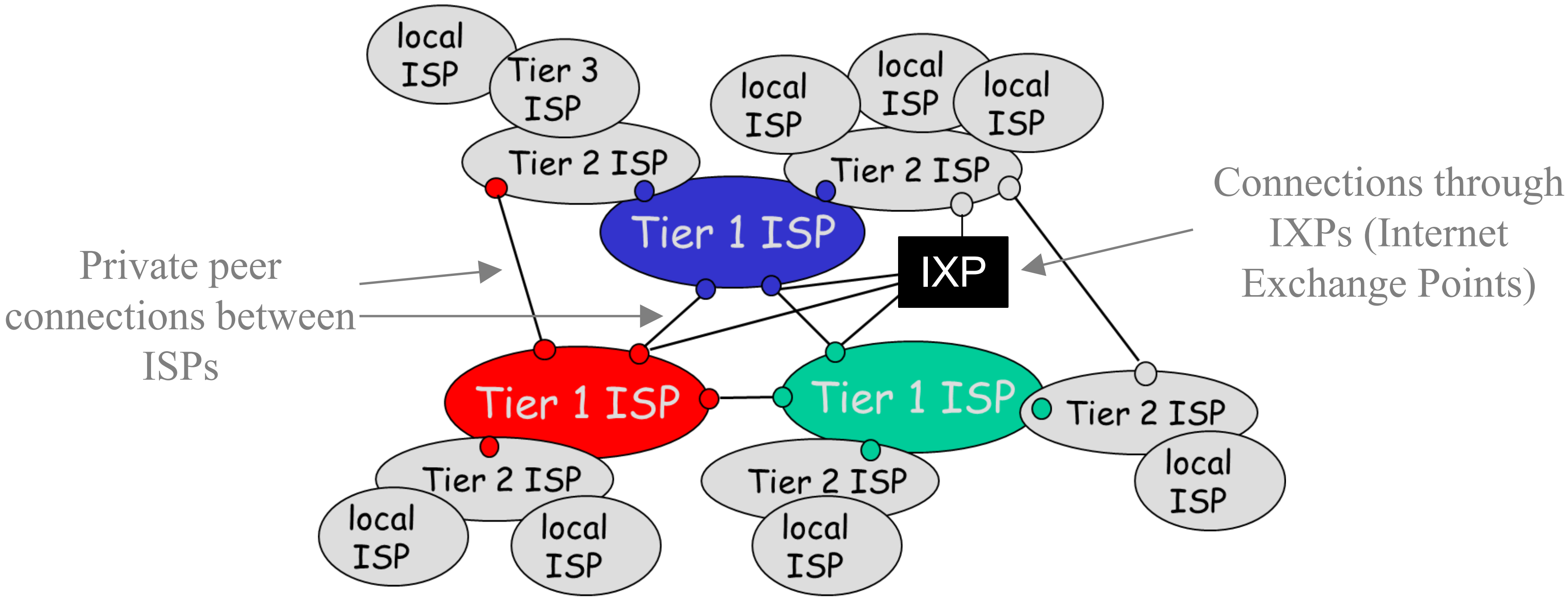
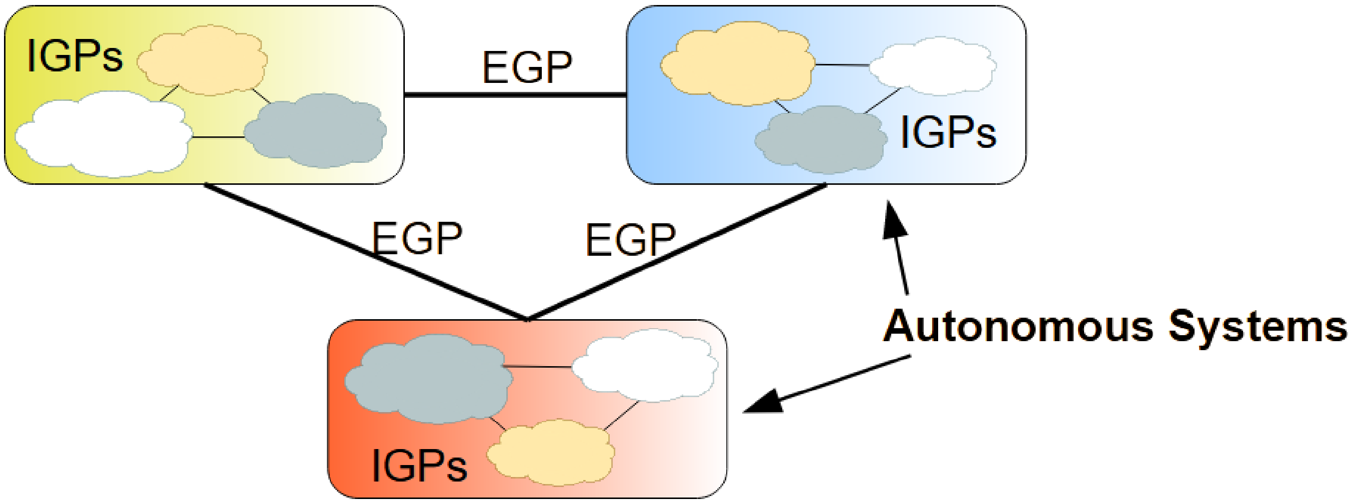
INTERNET: Network of Networks

- Aproximately hierarchical
  - Tier 1 ISPs (world-wide coverage)
    - treat themselves as equal
  - Tier 2 ISPs (regional coverage)
    - connect to Tier 1 ISPs (and possibly between them)
    - pay to Tier 1 ISPs for global connectivity
  - Tier 3 and local ISPs (providing Internet access to end users)
    - connect to Tier 2 ISPs
    - pay to Tier 2 ISPs for global connectivity



Border Gateway Protocol (BGP)



- The Border Gateway Protocol - version 4 (BGPv4) deployed in 1993 is currently the protocol that ensures Internet connectivity
- BGP is mainly used for routing between Autonomous Systems
- An Autonomous System (AS) is a set of networks under a single administration using IGP protocols (RIP, OSPF, IS-IS, etc.)

AS Identification

- An Autonomous System (AS) is identified by a number.
- AS numbers (ASNs) are assigned to Local Internet Registries (LIRs) and end-user organizations by their respective Regional Internet Registries (RIRs)
- RIRs, in turn, receive blocks of ASNs for reassignment from the Internet Assigned Numbers Authority (IANA).
- The IANA also maintains a registry of ASNs which are reserved for private use (and should therefore not be announced to the global Internet).

ASN (AS Number)

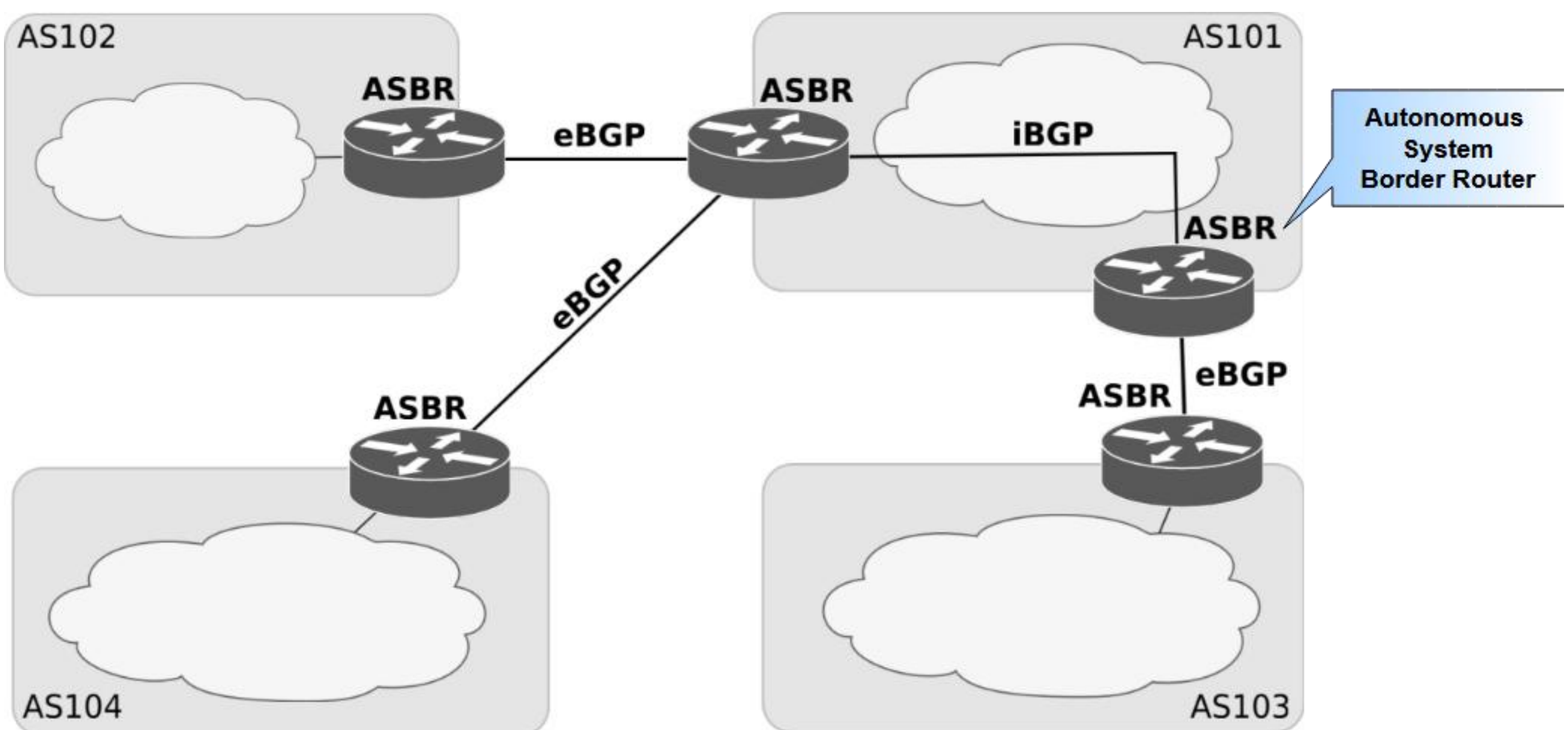
- RFC 4271 defines an ASN as a 2-bytes represented by a decimal
  - Private ASNs = 64512 through 65535
  - Public ASNs = 1 through 64511
  - 39000+ have already been allocated
  - Needed to expand ASN size from 2-bytes to 4-bytes
- RFC 4893 defines BGP support for 4-bytes ASNs
  - Maximum ASN value is 4,294,967,295
  - Since 2009 that all new assigned ASNs are 4-byte by default, unless otherwise requested.
  - The 4-byte ASNs are represented by two decimals (2-bytes each). Notation:
    - <higher2bytes in decimal>.<lower2bytes in decimal>
    - Example 1: AS 50000 is represented as "0.50000"
    - Example 2: AS 65536 is represented as "1.0"
    - Example 3: AS 65546 is represented as "1.10"

BGP Peer Relationship

- A BGP peer relationship is a connection between two routers that use BGP to exchange routing information
  - Usually configured by management
- Each peer relationship runs over TCP (port 179)
  - Ensures the reliable data delivery property of TCP
- BGP peers exchange all their routes in the form of UPDATE messages when the peer relationship is established
  - UPDATE messages are also sent to peers when there is a change in the known routes or a change in routing policy
- BGP peers exchange periodic KEEPALIVE messages
  - To check periodically the connectivity between peers (TCP does not exchange messages if no information is needed to be exchanged)
  - Low keepalive intervals can be set if fast fail-over is required

Internal BGP (iBGP) & External BGP (eBGP)

- Routers that implement BGP peer relationships are called Autonomous System Border Routers (ASBRs).
- BGP peer relationships can be established between:
  - ASBRs of the same AS (Internal BGP – iBGP)
  - ASBRs of different ASs (External BGP – eBGP)

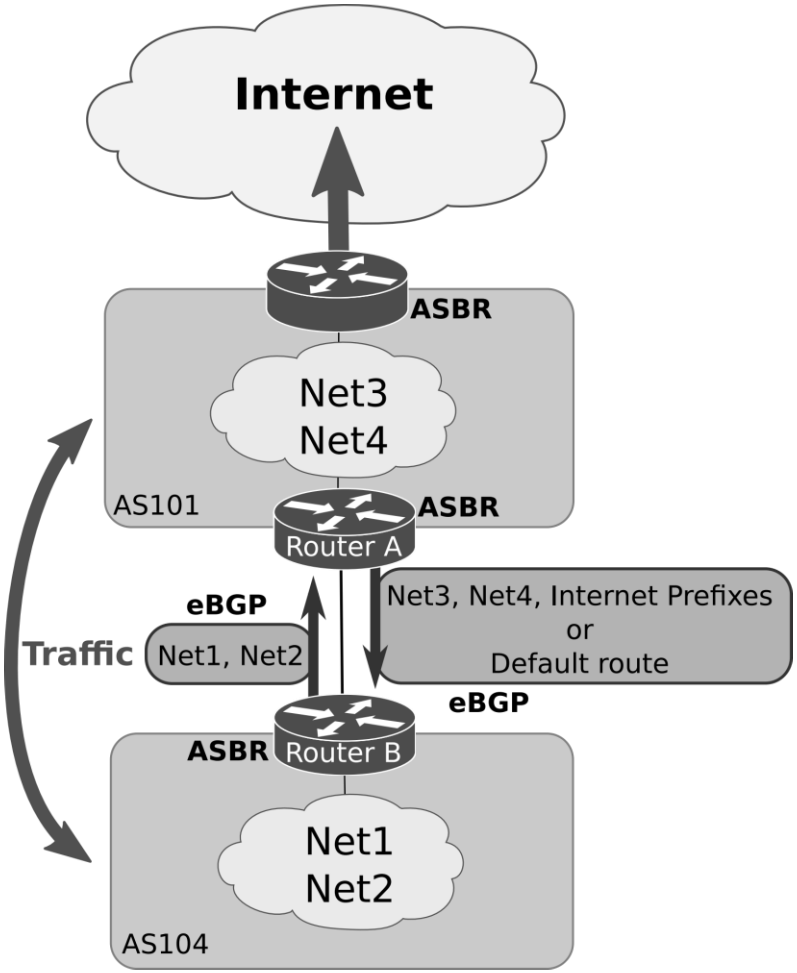


External and Internal BGP

- From the last slide:
  - External BGP (eBGP) is used between ASs.
  - Internal BGP (iBGP) is used within an AS.
- By default, a BGP router forwards the routes learned from one eBGP peer to both the other eBGP and iBGP peers.
  - Filters can be used to modify this behaviour.
- By default, a BGP router never forwards a route learned from one iBGP peer to another iBGP peer.
  - If can be differently if route-reflectors are used.
- **All ASBRs of an AS must maintain iBGP sessions with all other ASBRs in the same AS** (iBGP Mesh).
  - To obtain complete routing information from external routes.
  - Additional methods can be used to reduce iBGP Mesh complexity, like route reflectors and private ASs.

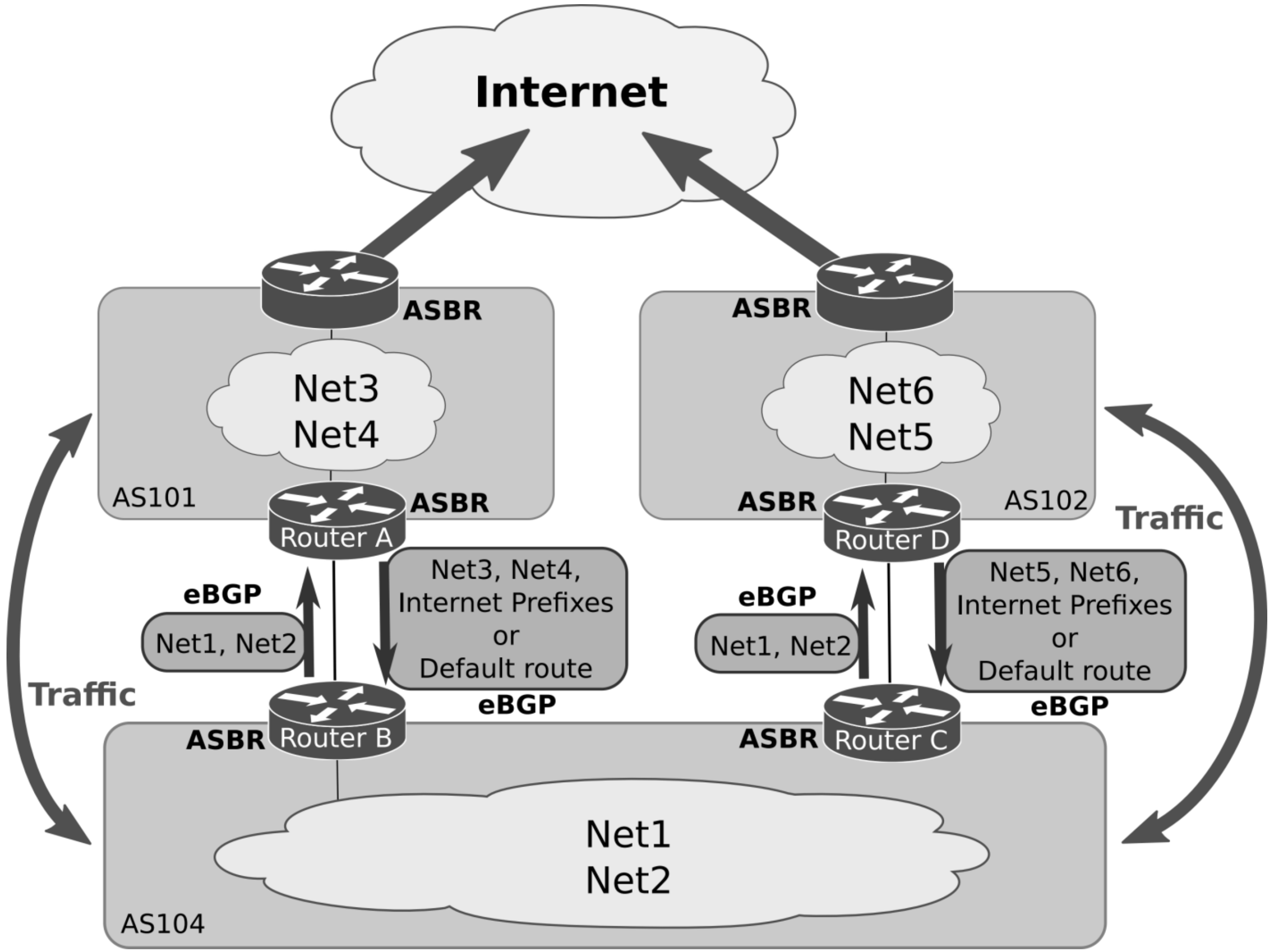
Types of Autonomous Systems: *Single-homed (or Stub) AS*

- AS has only one ASBR
  - Provides Internet access through one neighbour AS



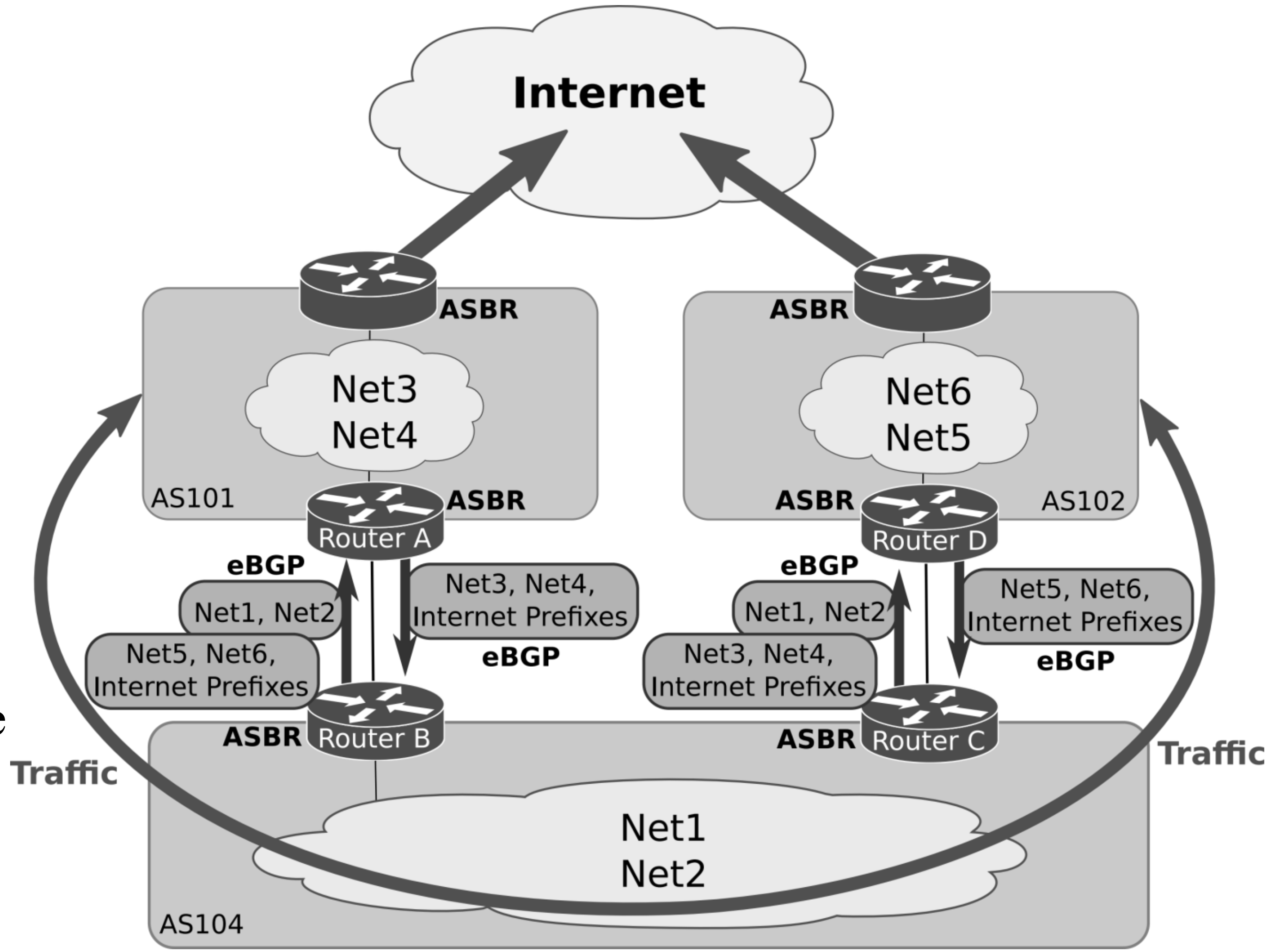
Types of Autonomous Systems: *Multi-homed Non-transit AS*

- AS has more than one ASBR
  - Provides Internet access through multiple ASs
- The AS does not support traffic between the neighbour ASs
  - by announcing only the internal routes to all neighbour ASs



Types of Autonomous Systems: *Multi-homed Transit AS*

- AS has more than one ASBR
  - Provides Internet access through multiple ASs
- The AS supports traffic between the neighbour ASs
  - by forwarding the routes learned from each neighbour AS to all other ASs

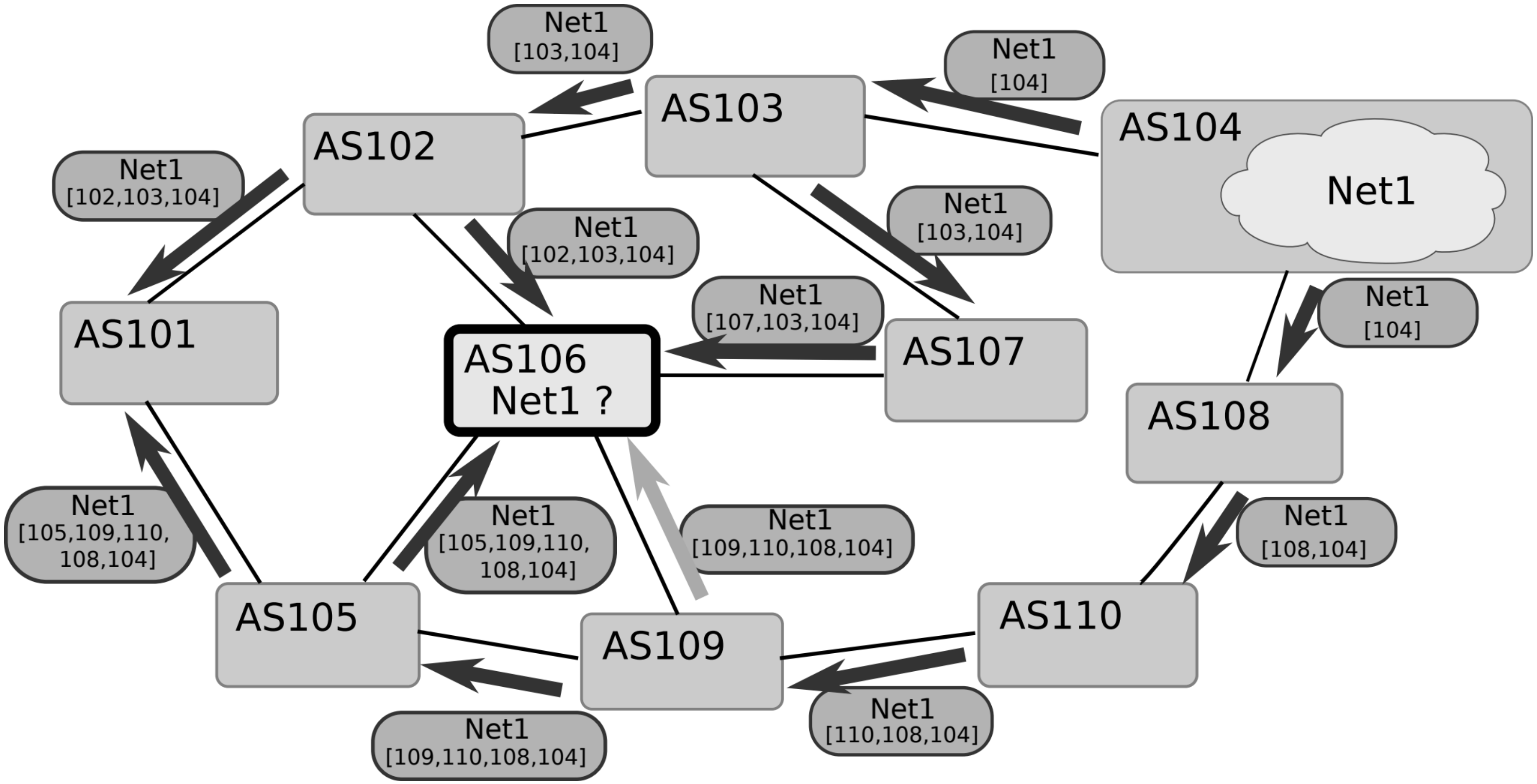


Path-Vector Protocol

- BGP is a path-vector protocol
- It works similarly to a distance-vector protocol, but each announced route contains the sequence of traversed ASs (identified by their ASNs)
  - Provides loop detection and policy-based routing selection
  - Illustration in the next slide
- An eBGP sending peer adds its own ASN to the AS sequence before forwarding the route announcement to its peer
- An iBGP sending peer does not modify the sequence because it is sending the route announcement to a peer within the same AS
  - The AS sequence list cannot be used to detect the iBGP routing loops

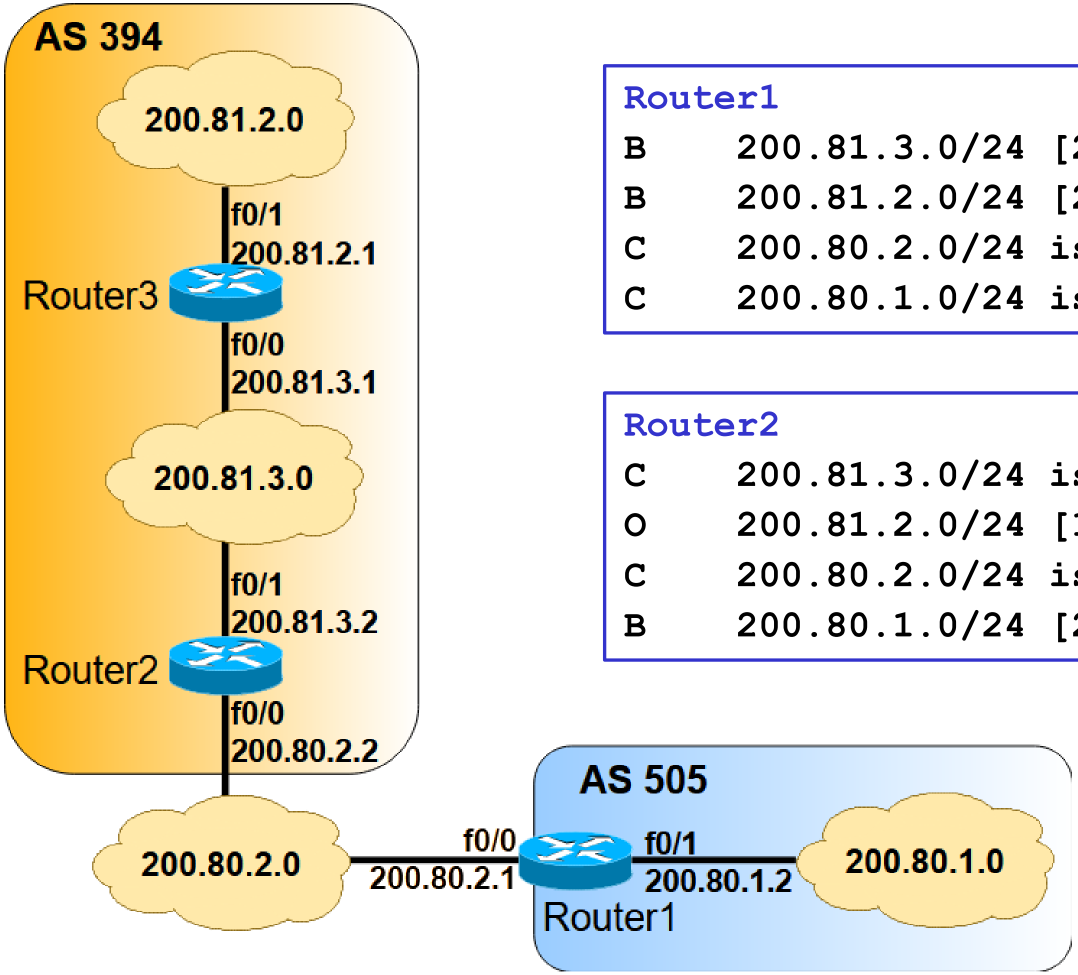
Path-Vector Protocol Illustration

- AS106 receives from its eBGP peers a route announcement of **Net1** (that belongs to AS104) with their routing AS sequence towards AS104
- Based on its routing configuration, AS106 selects the eBGP peer to forward the traffic towards **Net1**



Example

- Router1 is the ASBR of AS 505
- Router2 is the ASBR of AS 394
- eBGP session configured between Router1 and Router2
- AS 394 internal routing is configured with OSPF

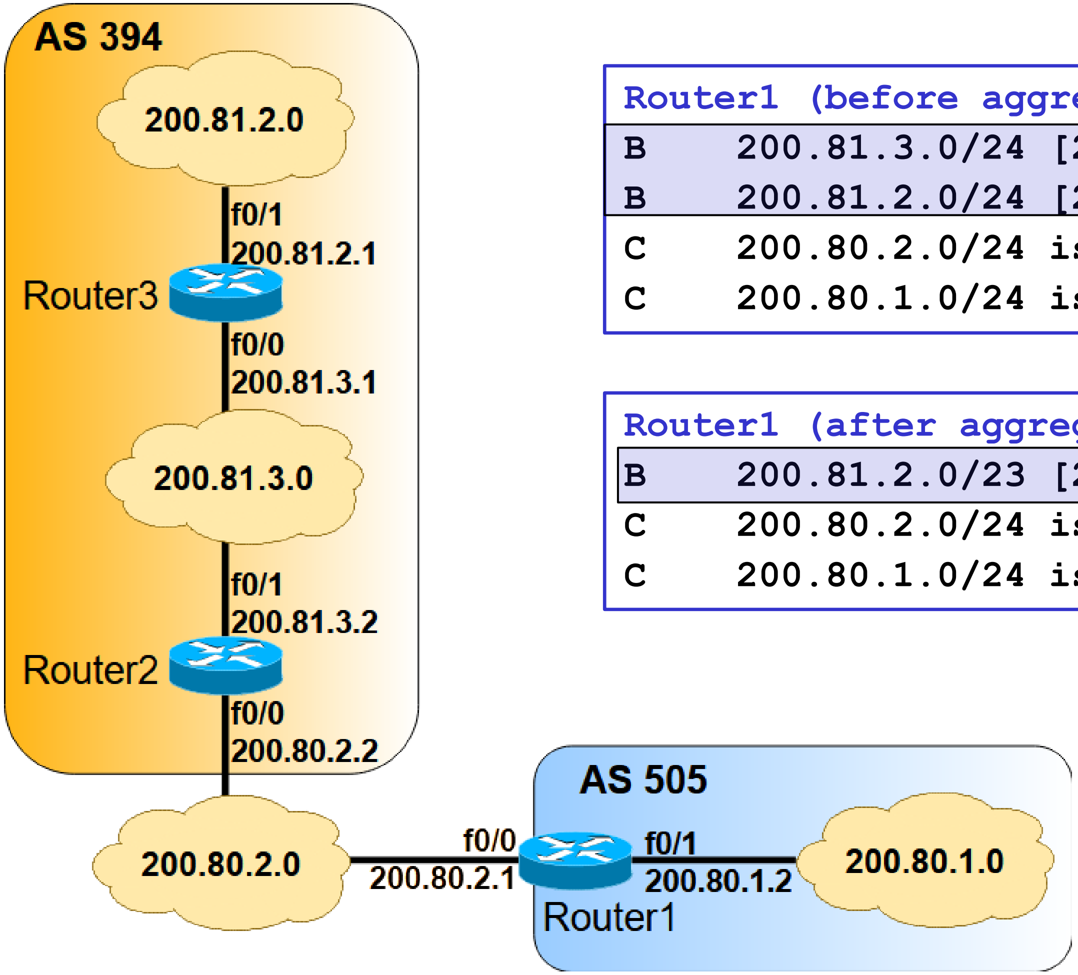



```
Router1
B     200.81.3.0/24 [20/0] via 200.80.2.2, 00:01:58
B     200.81.2.0/24 [20/0] via 200.80.2.2, 00:01:57
C     200.80.2.0/24 is directly connected, f0/0
C     200.80.1.0/24 is directly connected, f0/1
```

```
Router2
C     200.81.3.0/24 is directly connected, f0/1
O     200.81.2.0/24 [110/2] via 200.81.3.1, 00:01:12
C     200.80.2.0/24 is directly connected, f0/0
B     200.80.1.0/24 [20/0] via 200.80.2.1, 00:00:29
```

Example with an Aggregated Prefix

- 200.81.2.0/24 and 200.81.3.0/24 are aggregated by the prefix 200.81.2.0/23
- Aggregated prefix 200.81.2.0/23 is announced by Router2 to its eBGP peer Router1

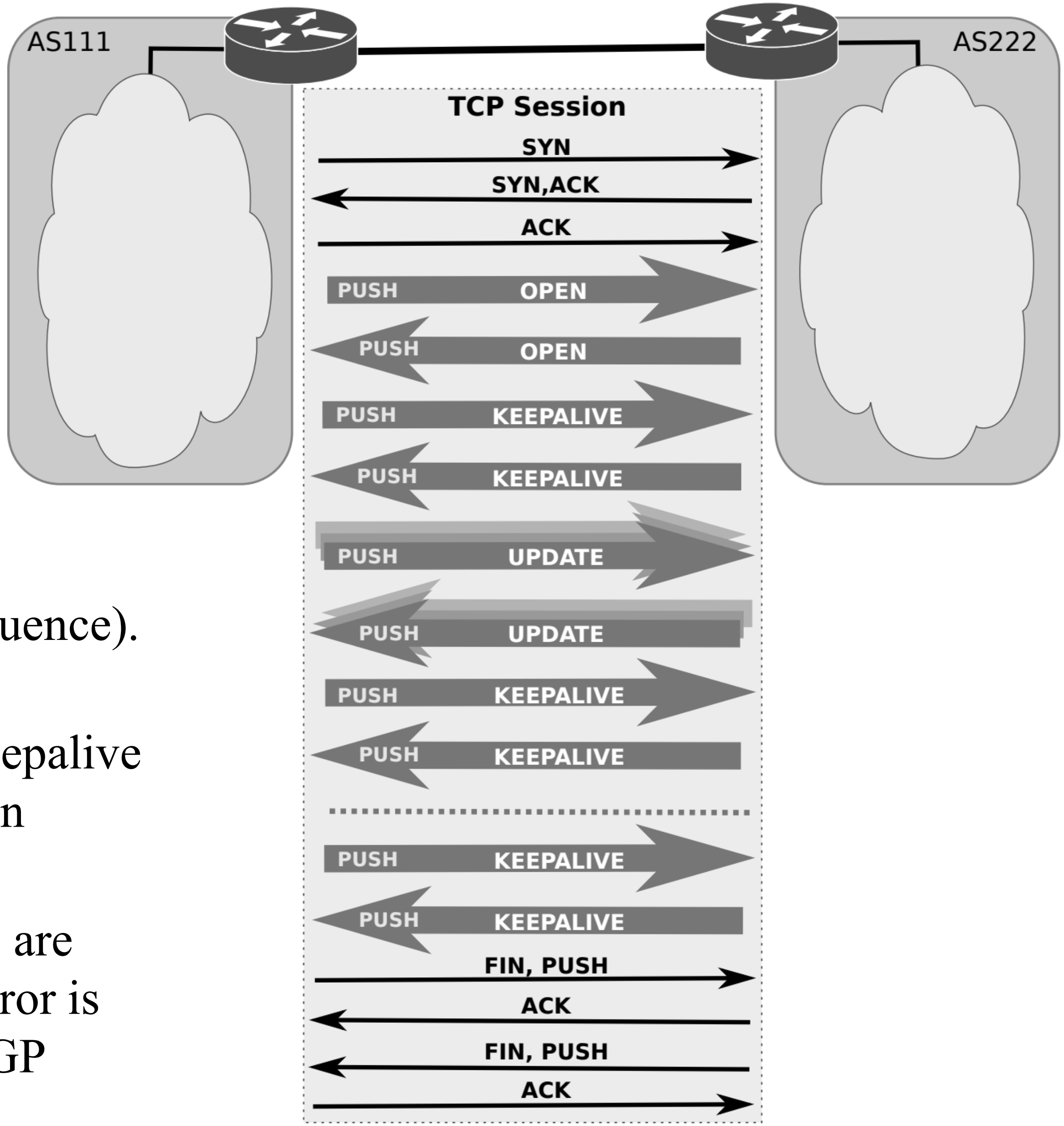



```
Router1 (before aggregation)
B     200.81.3.0/24 [20/0] via 200.80.2.2, 00:01:58
B     200.81.2.0/24 [20/0] via 200.80.2.2, 00:01:57
C     200.80.2.0/24 is directly connected, f0/0
C     200.80.1.0/24 is directly connected, f0/1
```

```
Router1 (after aggregation)
B     200.81.2.0/23 [20/0] via 200.80.2.2, 00:01:03
C     200.80.2.0/24 is directly connected, f0/0
C     200.80.1.0/24 is directly connected, f0/1
```

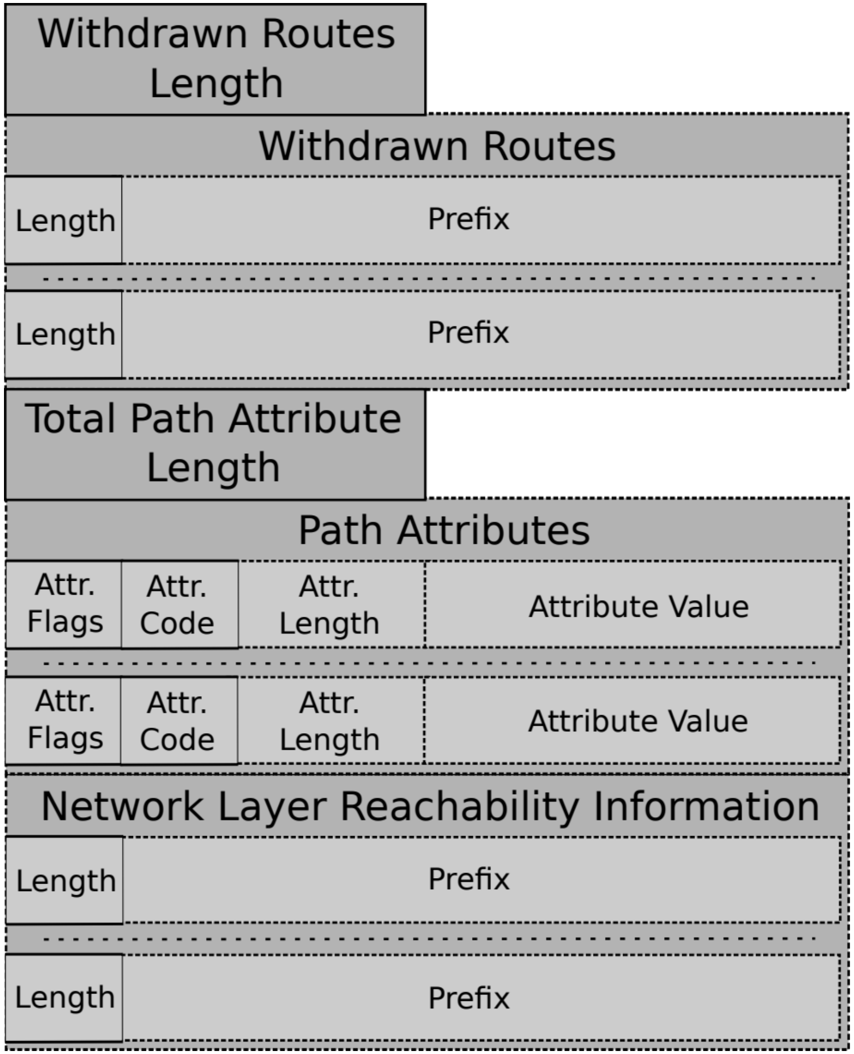
BGP Messages

- OPEN messages are used to establish the BGP session.
- UPDATE messages are used to send IP prefixes, along with their associated BGP attributes (such as the AS_PATH with the AS sequence).
- KEEPALIVE messages are exchanged whenever the keepalive period is reached, without an update being exchanged.
- NOTIFICATION messages are sent whenever a protocol error is detected, after which the BGP session is closed.



BGP UPDATE Message Format

- Withdrawn Routes
  - list of IP prefixes no longer accessible.
- Path Attributes
  - attributes associated with the IP prefixes (listed next) that are used to compute the best routing path.
- Network Layer Reachability Information
  - list of known IP prefixes characterized by the attributes specified in the previous Path Attributes.



BGP Attributes

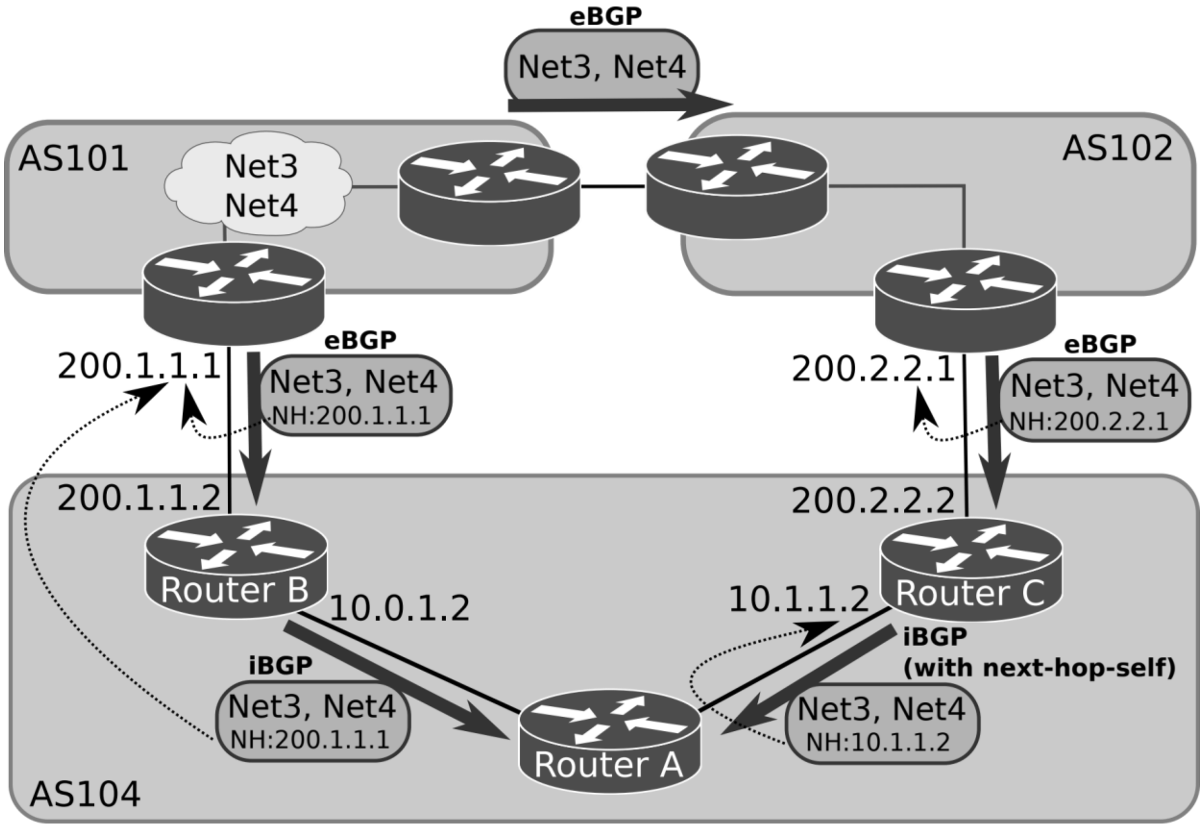
- BGP attributes (included in the BGP UPDATE messages) are metrics describing the characteristics of BGP routes towards the announced IP prefixes and are used to select routing paths.
- There are different attribute categories:
  - Well-known Mandatory (always included in BGP UPDATEs)
    - AS_PATH, ORIGIN, Next-Hop
  - Well-known Discretionary (may or may not be included in BGP UPDATEs)
    - Local Preference, Atomic_Aggregate
  - Optional Transitive (may not be supported by all BGP implementations)
    - If not supported, BGP routers forward them unchanged.
    - Aggregator, Community, AS4_Aggregator, AS4_PATH
  - Optional Non-transitive (may not be supported by all BGP implementations)
    - Multi-exit-discriminator (MED), Originator, Cluster-ID
  - Cisco-defined (local to router, not included in BGP UPDATEs)
    - Weight

BGP Well-known Mandatory Attributes: AS_PATH and ORIGIN

- AS_PATH
  - When a route advertisement passes through an Autonomous System, the AS number is added to an ordered list of AS numbers that the route advertisement has traversed (the basis of the path-vector property).
- ORIGIN
  - Indicates how BGP has learned the route in the originating AS. It can take one of three possible values:
    - IGP (value 0) is used if the route is interior to the originating AS, resulting from an explicit inclusion of the network prefix within the BGP routing process.
    - EGP (value 1) is used if the route is learned by other EGP protocol (no longer used in current IP networks).
    - INCOMPLETE (value 2) is used if the route is learned by other means, namely, route redistribution from other routing protocols into the BGP routing process.

BGP Well-known Mandatory Attribute: Next-Hop

- The eBGP Next-Hop attribute is the IP address used to reach the advertising router, i.e., the IP address of the eBGP peer.
  - By default, the IP networks of eBGP sessions do not belong to any AS.
  - So, the IP address of the eBGP Next-Hop attribute is not known inside an AS.
- In the BGP router, the Next-Hop attribute announced to its iBGP neighbours must be configured with its IP address inside the AS.
  - See example of Router C in the figure.

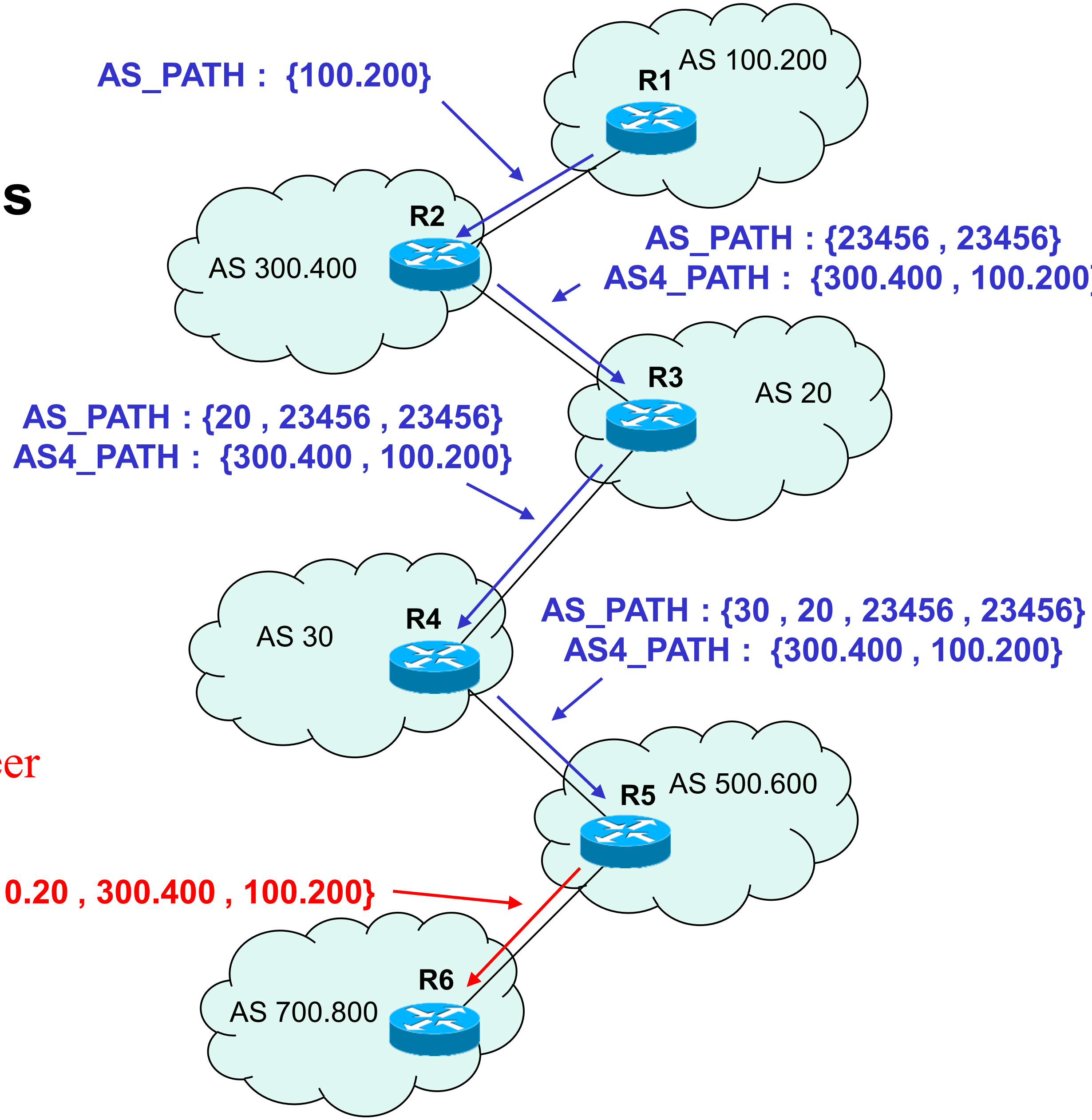


BGP Optional Transitive Attribute: AS4_PATH

- AS4_PATH attribute is used to make compatible the routing between Autonomous Systems supporting 4-bytes ASNs and Autonomous Systems not supporting 4-bytes ASNs
- AS4_PATH attribute has the same semantics as the AS_PATH attribute, except that it is Optional Transitive, and it carries an ordered list of 4-bytes ASNs.
  - 4-byte AS support is advertised via BGP capability negotiation in the initial exchanged OPEN messages
  - Peers supporting 4-byte Autonomous Systems are known as NEW BGP peers and process incoming AS4_PATH attributes
  - Peers not supporting 4-byte Autonomous Systems are known as OLD BGP peers and forward AS4_PATH attributes unchanged
- New Reserved 2-byte ASN: AS_TRANS = 23456
  - 2-byte placeholder for a 4-byte ASN in the AS_PATH attribute
  - Used for backward compatibility between OLD and NEW BGP speakers

Illustration of AS_PATH and AS4_PATH Attributes

- R3 and R4 are OLD BGP peers
- Since AS4_PATH is an Optional Transitive attribute, OLD BGP peers forward them unchanged
- R5 merge AS_PATH and AS4_PATH to the next peer

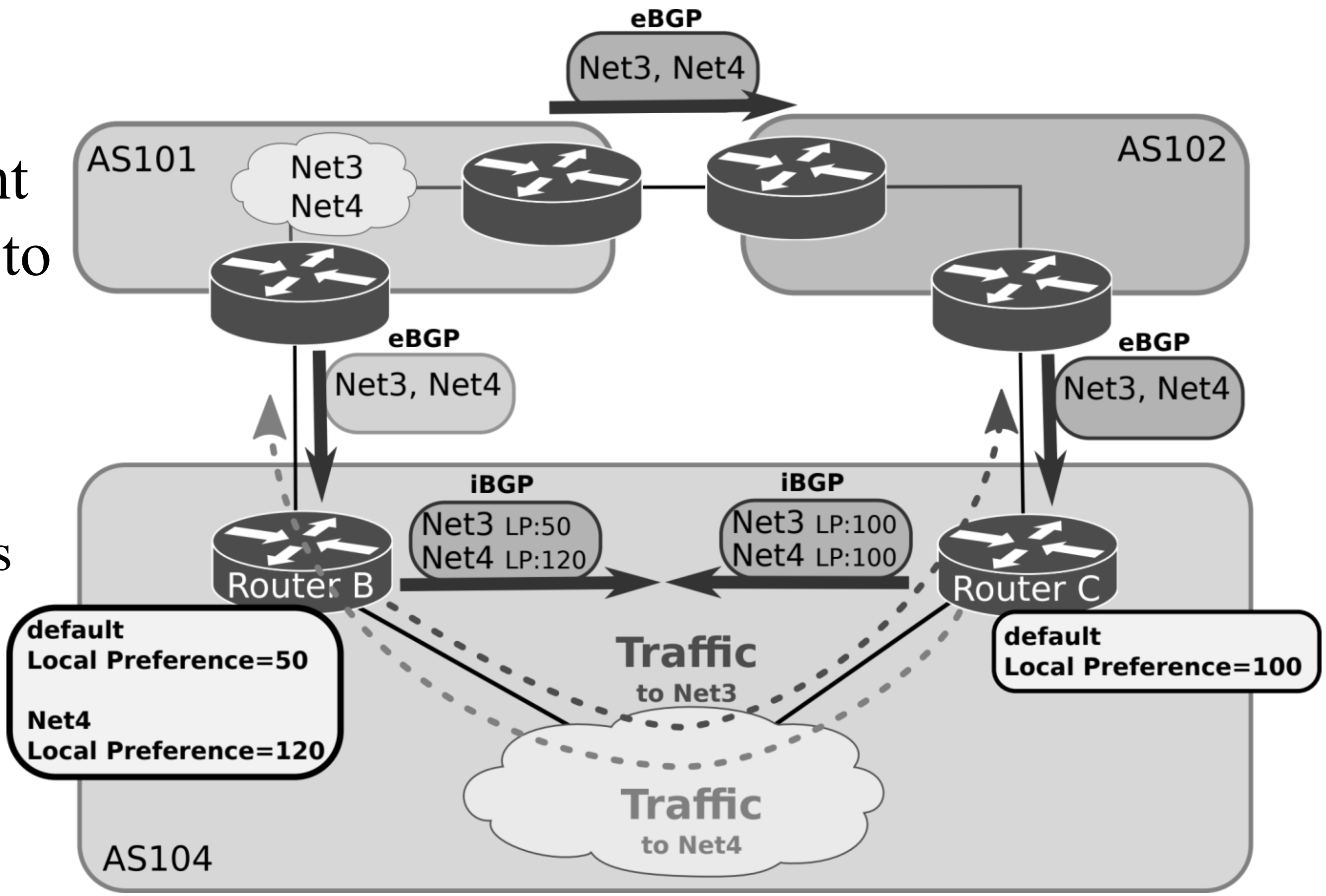


Processing of AS_PATH and AS4_PATH in NEW BGP routers

- A NEW BGP router sends UPDATEs to a NEW BGP peer of another AS
  - It sends AS_PATH with 4-bytes ASNs adding its own ASN
- A NEW BGP router sends UPDATEs to an OLD BGP peer of another AS
  - It sends AS_PATH with 2-bytes ASNs adding its own ASN and replacing 4-bytes ASNs by the AS_TRANS value
  - It sends AS4_PATH with 4-bytes ASNs adding its own ASN
- A NEW BGP router receives UPDATEs from a NEW BGP peer
  - It decodes each ASN as 4-bytes in the AS_PATH attribute
- A NEW BGP router receives UPDATEs from an OLD BGP peer
  - It determines the correct AS_PATH by merging the received AS_PATH and AS4_PATH attributes. Example:
    - Received AS_PATH : { 250 , 225 , 23456 , 23456 , 200 , 23456 , 175 }
    - Received AS4_PATH : { 100.1 , 100.2 , 0.200 , 100.3 , 0.175 }
    - Merged AS-PATH : { 0.250 , 0.225 , 100.1 , 100.2 , 0.200 , 100.3 , 0.175 }

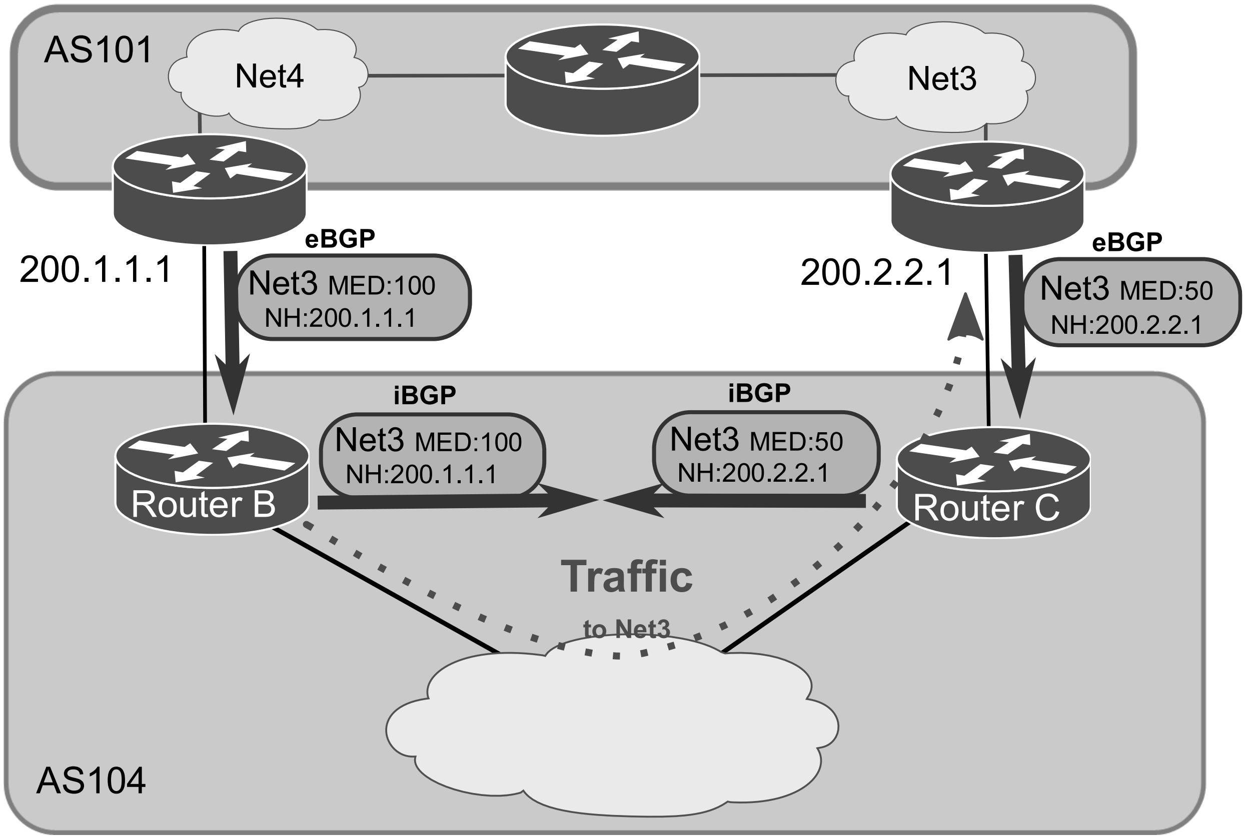
BGP Well-known Discretionary Attribute: Local Preference

- Local Preference (LP) is used to select the exit point from the local AS to outside routes.
  - **Higher value** is preferred.
  - The LP attribute is only propagated in the local AS.
  - The LP value can be different, for different routes.



BGP Optional Non-transitive Attribute: Multi-Exit Discriminator (MED)

- MED is used as a suggestion to a peer AS, aiming to influence the incoming entry point traffic.
  - **Lower value** is preferred.
  - The external AS receiving the MED values may use other BGP attributes for route selection.



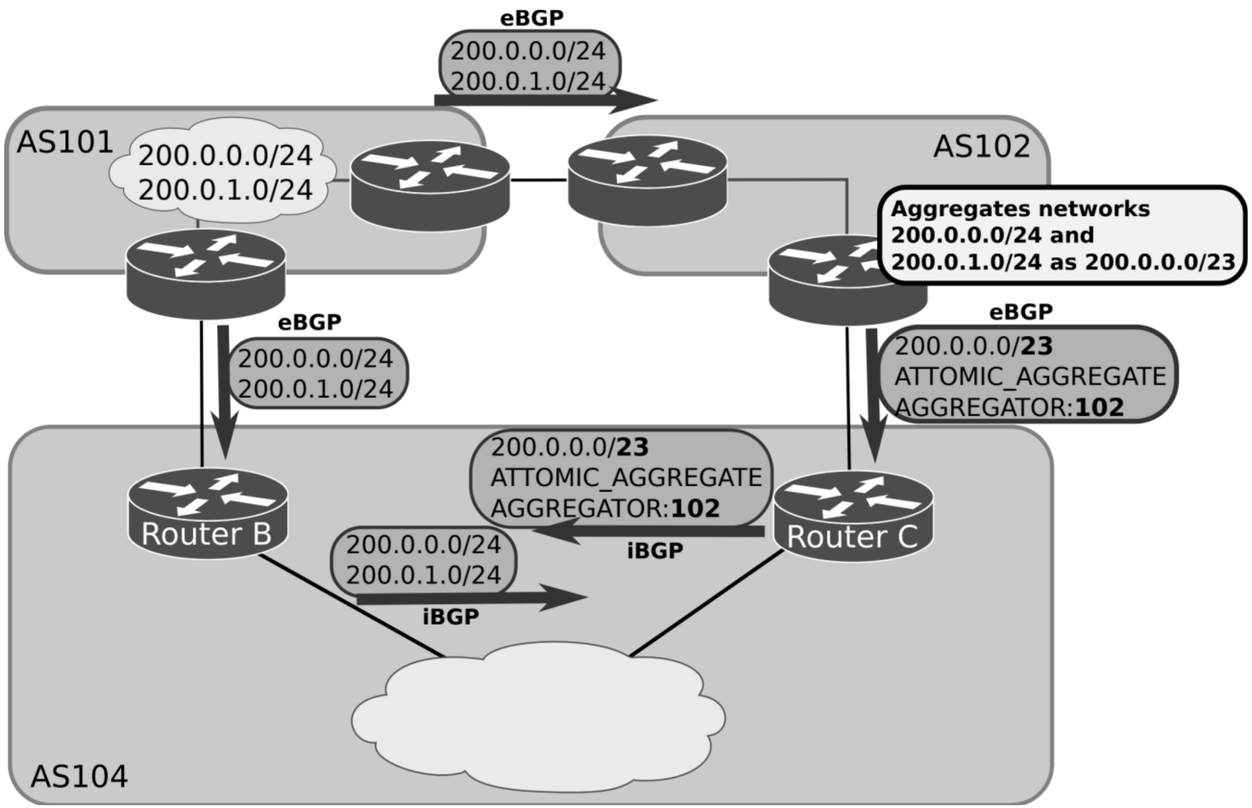
Atomic_Aggregate and Agregator Attributes

Atomic_Aggregate (Well-known Discretionary attribute)

- It indicates that the announced route is an aggregation of more specific routes.
- Information of the specific routes is lost.

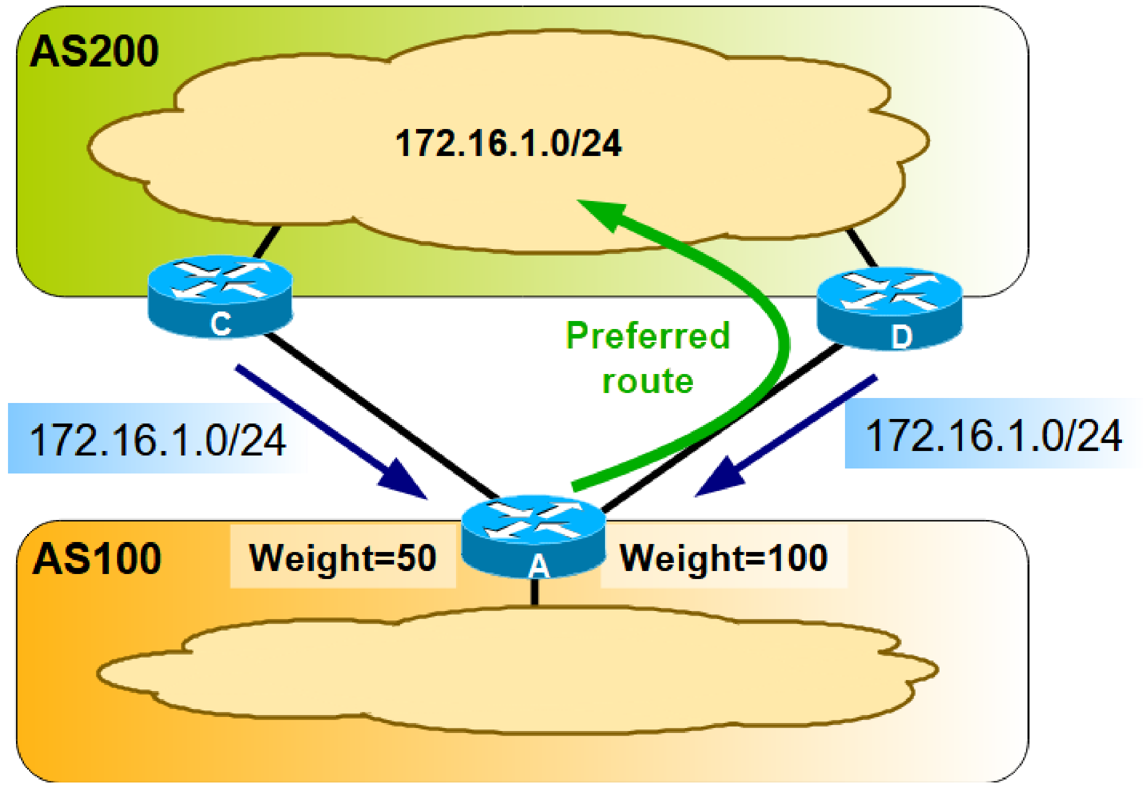
Aggregator (Optional Transitive attribute)

- It provides information about which AS performed the aggregation.
- It provides also the Router ID of the BGP router that originated the aggregate.



CISCO Weight Attribute

- Weight is a Cisco-defined local attribute.
  - The weight attribute is not advertised to BGP peers.
  - It is useful when a BGP router has multiple eBGP peers.
- If the router learns more than one route to the same destination prefix, the route with **the highest weight** is preferred.

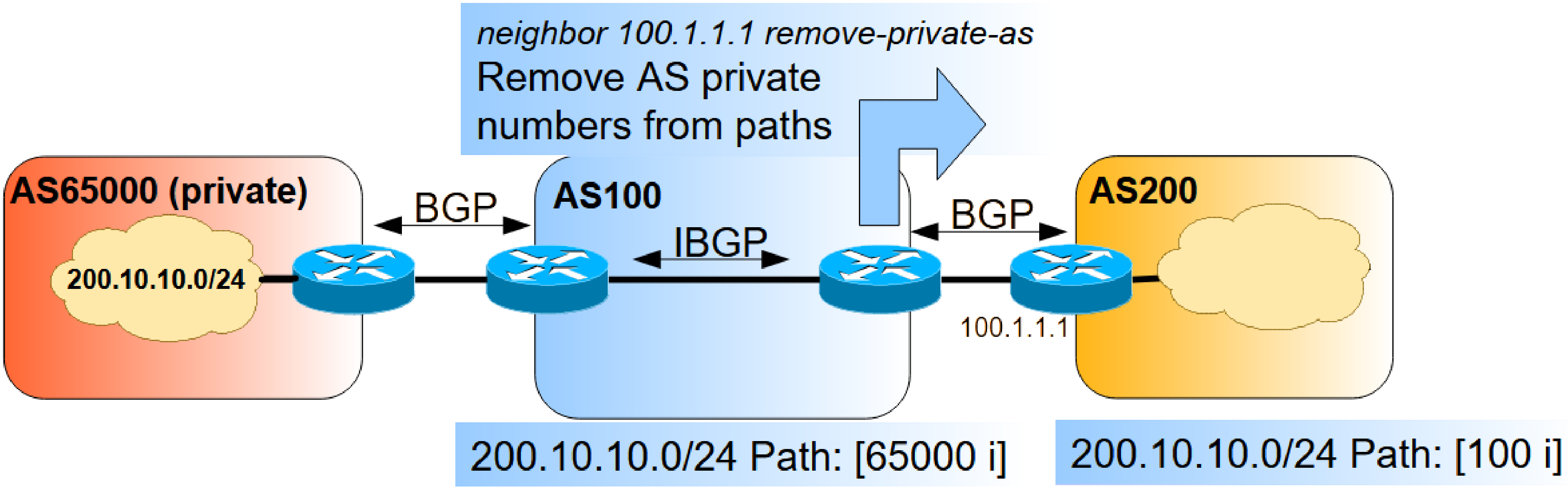


BGP Path Selection

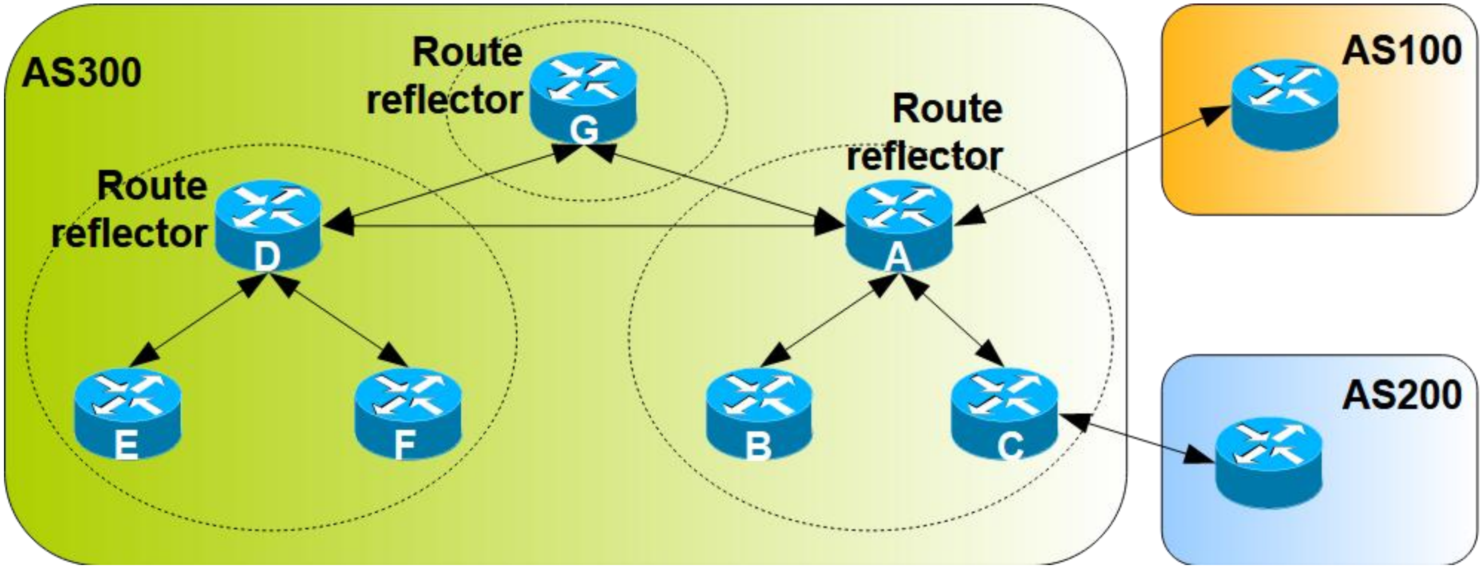
- BGP may receive advertisements for the same network prefix from multiple neighbour routers and selects only one path.
- BGP sets the selected path in the IP routing table of the router and propagates the path to its BGP neighbours.
- The path selection uses the following criteria, in the order:
  - Largest weight value (Cisco only)
  - Largest Local Preference value
  - Path that was originated locally (i.e., belongs to the local AS)
  - Shortest path (i.e., with a shorter AS_PATH attribute length)
  - Lowest ORIGIN value
  - Lowest MED value
  - External path (through eBGP) over internal path (through iBGP)
  - Lowest IGP Metric (the path with the lowest IGP cost to the Next Hop)
  - Oldest announced by all neighbour BGP routers
  - Lowest neighbour BGP Router ID

BGP Private Autonomous System (AS)

- When a customer network is large, the network might be set as an Autonomous System with a given ASN:
  - Assigning a **Public** ASN
    - Required when the customer network connects to different ISPs
  - Assigning a **Private** ASN (for example, in the range of 64512 to 65535)
    - Possible when the costumer networks connects to a single ISP (not recommended if the costumer plans to connect to other ISPs in the future)

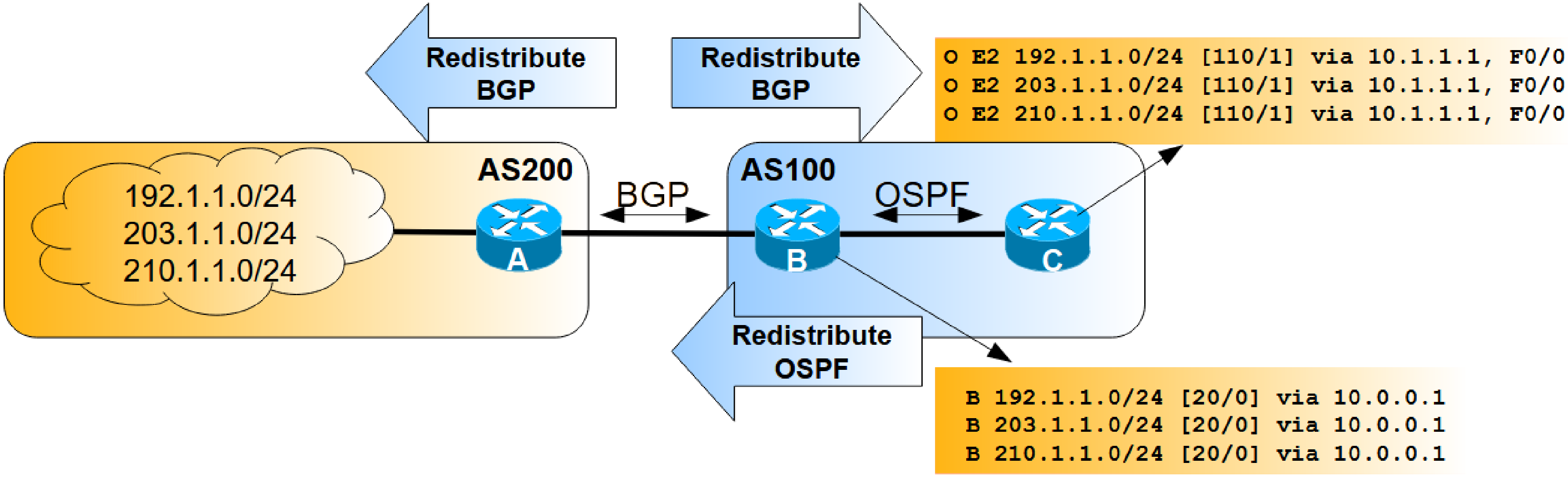


BGP Route Reflectors



- Without route reflectors, the network requires a full iBGP mesh within AS300 in the above example.
- The route reflector and its clients are called a cluster. In the figure:
  - One cluster with Router A configured as a route reflector with clients B and C (iBGP peering between Routers B and C is not required).
  - One cluster with Router D configured as a route reflector with clients E and F (iBGP peering between Routers E and F is not required).
- Route reflector routers are connected in a full iBGP mesh.

Routes Redistribution



- Redistributing IGP routes by BGP:
  - Simplify BGP configuration (advantage)
  - BGP announces to outside only internal networks with connectivity (advantage)
  - UPDATEs must be exchanged when the internal networks change (disadvantage)
- Redistributing BGP routes by IGP protocols:
  - Internal routers know all external routes without the need of default routes (advantage)
  - Increases routing tables size in internal routers (performance disadvantage)

BGP Filtering

- By default, BGP announces every network path received from other BGP peers or redistributed by IGP.
- Sending and accepting BGP updates can be controlled by using different filtering methods.
  - Route-maps, prefix-lists, distribution-lists.
- BGP updates can be filtered based on:
  - Route information, Path information (AS_PATH attribute), etc…
- Most common actions:
  - Configure Local Preference values to define the preferred paths.
  - Block/Deny paths to prevent undesired routing paths of being selected.
- Best basic practices:
  - Block all IPv4 private networks,
  - Announce default routes only to AS peers with a traffic transport contract.
  - Accept default routes only from AS peers providing a traffic transport service.

Multi-Protocol Border Gateway Protocol (MP-BGP)

- BGP carries only routing information of IPv4 unicast
- MP-BGP is an extension to the BGP protocol:
  - It behaves as BGP for IPv4 unicast and, therefore, a BGP peer relationship is possible between a BGP router and a MP-BGP router.
- MP-BGP exchanges Multi-Protocol NLRI (Network Layer Reachability Information)
- MP-BGP carries routing information about different protocols/families:
  - IPv4 and IPv6 (both Unicast and Multicast)
  - Multi-Protocol Label Switching (MPLS) VPNs (IPv4 and IPv6)
  - 6PE: IPv6 packets transported over an IPv4 MPLS backbone
  - 6VPE: Multiple IPv6 VPNs created over an IPv4 MPLS backbone
  - Ethernet VPN (EVPN): a protocol to distribute IP and MAC addresses
    - it is a standards-based solution for VXLANs used in Data Centres

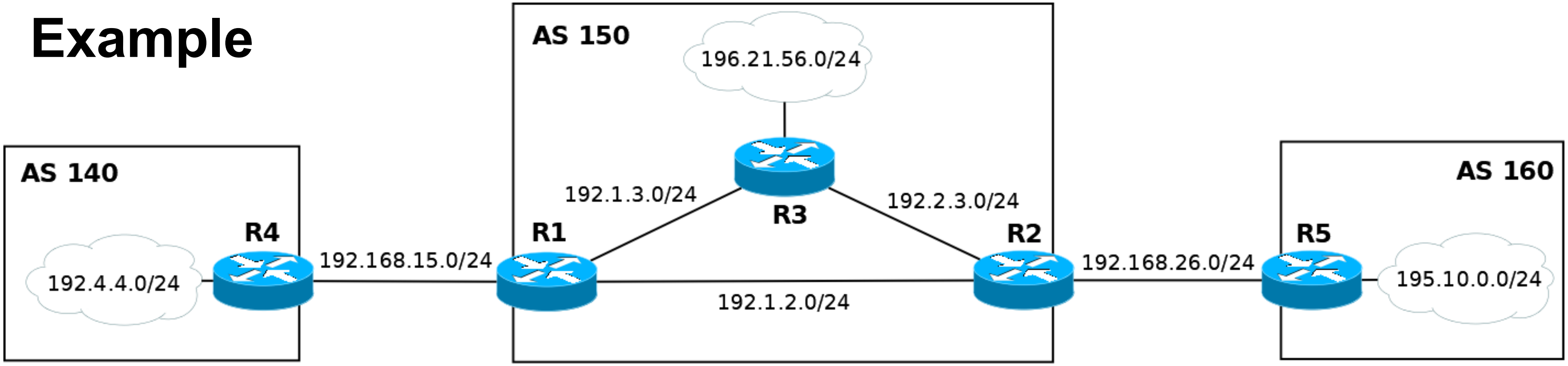
MP-BGP New Attributes

- MP-BGP introduces new non-transitive and optional attributes:
  - MP_REACH_NLRI
    - Carry the set of reachable destinations together with the next- hop information to be used for forwarding to these destinations
  - MP_UNREACH_NLRI
    - Carry the set of unreachable destinations
- The new attributes contain fields to identify the different protocols/families
  - Address Family Identifier (AFI) with SAFI (Subsequent AFI)
    - AFI is IPv4 or IPv6, for example
    - In the case IPv4 or IPv6, SAFI is unicast or multicast
    - AFI / SAFI provide the protocol information associated with the content of the NLRI (Network Layer Reachability Information)
  - Next-Hop
    - Next-hop address must be of the family identified by the AFI

MP-BGP Negotiation Capabilities

- Like BGP, MP-BGP routers establish BGP peer relationships through the exchange of OPEN messages
  - MP-BGP introduces a new optional parameter: CAPABILITIES
- MP-BGP negotiation is the process of determining the supported capabilities in a BGP peer relationship:
  - process done by identifying the CAPABILITIES parameters announced in the OPEN messages that are common to both BGP peers.
- Some important CAPABILITIES parameters:
  - Multi-Protocol extensions (AFI/SAFI): identify the supported AFI/SAFI protocol/families
  - Route Refresh: allows a BGP router to request the resend of the reachability information from the BGP peer router
  - Outbound Route Filtering (ORF): allows a BGP router to send (to the BGP peer router) prefix information that will be denied if announced in the inbound direction

Example



Consider the IPv4 network above with 3 Autonomous Systems (the hostID of the IPv4 addresses on each router is the number of the router name). Routing inside AS 150 is configured with OSPF (all interfaces have a cost of 1) and with a default route of type E2 in R1. In AS 150, the OSPF routes are redistributed by BGP but the BGP routes are not redistributed by OSPF. All required BGP peer connections are properly configured to have full IP connectivity between all Autonomous Systems.

a) What are the AS_PATH and NEXT HOP attributes of prefix 195.10.0.0/24 announced in an UPDATE message sent by R1 to R4?

b) What are the AS_PATH and NEXT HOP attributes of prefix 192.4.4.0/24 announced in an UPDATE message sent by R1 to R2?

c) Indicate (and justify) the path of the ICMP Echo Request messages of a ping from a host in 196.21.56.0/24 to a host in 195.10.0.0/24.

d) Explain if a ping from a host in 196.21.56.0/24 to the address 192.168.15.4 is or in not successful.